	Target engagement	Dose	Efficacy	Safety
#7	**Alzheimer disease** anti-MAPT antibodies (4)	**2023** Gosuranemab immunodepletion of AD/PSP brain… △ Tau captured; CSF Abeta42 flat	**2022** P301L/P301S tau-transgenic mice (JNPL3, P301S, P301L, rTg4510)	**2021** Human AD-brain PHF-tau seeding in HEK293 biosensor cells △ Four mouse successes, four…	**2021** Cynomolgus and tau-transgenic mice (PK/PD) Model data unpublished
	Alzheimer disease HRH3 antagonists (3)	**2012** Rodent procognitive battery with exposure-response and ex-viv… Not measured in patients	**2012** Rodent procognitive battery with exposure-response and ex-vivo H3 occupancy (rat inhibitory…	**2014** Rodent procognitive battery… Three rodent successes, three AD failures	**2012** Rat safety pharmacology and sleep-wake EEG (with monkey… △ Sponsor confirms dose-depend…
	Alzheimer disease HTR6 antagonists (3)	**2014** Rat in-vivo striatal 5-HT6 occupancy ([3H]Lu AE60157) and… △ Rat under-called brain penetration	**2022** Rat cognition battery (object recognition, water maze, radial arm maze) and hippocampal ACh…	**2019** Rat cognition battery (object… Rat efficacy did not translate; Phase 2 null	**2019** IND-enabling toxicology (unpublished) Model data unpublished
	Alzheimer disease anti-APP antibodies (3)	**2018** tg-ArcSwe mice, mAb158 Mice: no plaque fall; patients: PET fell	**2012** hAPP/PS1 mice, crenezumab No checkable prediction	**2022** In-vitro binding to Aβ monomers, protofibrils and fibrils (inhibition ELISA, immunodepletio…	**2017** Aged APP mice given 2H6 (ponezumab parent) △ Excess microbleeds missed
#8	**asthma** PTGDR2 antagonists (9)	**2016** Human eosinophils and Th2 cells in vitro	**2016** Human eosinophils and Th2 cells in vitro	**2011** Allergen challenge	**2012** Off-target selectivity panels; animal toxicology unpublished Model data unpublished
	asthma ADRB2 agonists (4)	**2017** Carmoterol binding kinetics… Peak FEV1 up, as tight binding implied	**2014** Guinea-pig, human bronchi, dog bronchoprotection (abediterol) △ 2 ug too low; 4 ug worked	**2014** Guinea-pig/human bronchial onset and duration (AZD3199)	**2017** Guinea-pig atria and papillary muscle chronotropy/inotropy, cardiac side-effect model (carmoterol)
	asthma NR3C1 agonists (4)	**2017** Rat allergic airway model: lung oedema, thymic involution FeNO fell at all AZD7594 doses	**2009** Rat allergic airway model: lung oedema, thymic involution	**2015** Late asthmatic response to allergen in mild atopic asthmatics Not measured in patients	**2020** Rat allergic airway model: lung oedema, thymic involution
	asthma anti-IL13 antibodies (4)	**2018** IL-13-stimulated bronchial epithelial cells and patient brushings △ Airway eosinophils did not fall	**2017** Tralokinumab population PK/PD dose-exposure-FEV1 model	**2017** Whole-lung allergen challenge in mild allergic asthmatic… Second ICS-free FEV1 trial missed	**2023** Cynomolgus repeat-dose… Clean monkey tox; LAVOLTA safety placebo-like
#9	**breast cancer** CDK4/6 inhibitors (6)	**2013** Human tumour xenografts in mice (phospho-Rb Ser780)	**2017** Rodent brain penetration and intracranial xenograft (abemaciclib)	**2020** Human breast cancer cell-li… PALLAS adjuvant palbociclib fails iDFS	**2017** Dog cardiovascular telemetry (QTc) with hERG assays △ Missed liver injury and clots
	breast cancer ESR1 degrader (5)	**2016** ESR1-mutant MCF7-Y537S cells, Y537S and mutant-ER PDX	**2026** ESR1-mutant MCF7-Y537S cells, Y537S and mutant-ER PDX	**2018** ER+ ESR1-wild-type breast xenografts (MCF7, tamoxifen-resistant MCF7, HCI013…	**2026** Zebrafish embryo heart-rate model of SERD-induced bradycardia (wild-type and esr1/esr2/gper…
	breast cancer HSP90AA1/HSP90AB1 inhibitors (5)	**2014** HER2+ SKOV-3 xenograft, 89Zr-trastuzumab PET of luminespib HER2 downregulation	**2012** Rat and dog repeat-dose toxicology (alvespimycin) No checkable prediction	**2013** HER2+ breast xenografts HER2+ trial: 0 of 26 responses, stopped	**2013** Mouse xenografts, cardiac… No cardiac or liver signal; GI unpredicted
	breast cancer PI3K/MTOR inhibitors (5)	**2018** BEZ235 xenografts, tumour PI3K/Akt readout	**2018** BEZ235 xenografts, tumour PI3K/Akt readout	**2015** XL765 xenografts, pAKT PD No responses, PFS6 8% in phase II	**2013** BEZ235 xenografts, tumour… Well-tolerated claim broken by GI harm in breast
	breast cancer anti-IGF1R antibodies (4)	**2011** MCF-7 mice, IMC-A12	**2016** MCF-7/AC-1 xenografts under letrozole No checkable prediction	**2017** Tamoxifen-resistant MCF-7L cells and xenografts (dalotuzumab) △ No efficacy in HR+ breast cancer	**2016** MCF-7/AC-1 xenografts under letrozole
	breast cancer PARP1/2 inhibitors (3)	**2009** MX-1 BRCA1/2-deficient breast xenograft and syngeneic models, veliparib (ABT-888)…	**2009** MX-1 BRCA1/2-deficient breast xenograft and syngeneic models, veliparib (ABT-888)…	**2019** BRCA1-deficient K14cre;Brca1F/F;p53F/F… △ Veliparib + carboplatin: nothing	**2020** 2X-121 BRCA-deficient mice, PAR PD, chemo potentiation △ Anaemia, neutropenia unforeseen
	breast cancer PI3K (pan) inhibitors (3)	**2011** PI3K-activated lines and mice	**2014** PI3K-activated lines and mice	**2015** PI3K-activated lines and mice △ PIK3CA-mutant enrichment not…	**2015** PI3K-activated lines and mice Liver toxicity unflagged by preclinical package
	breast cancer PIK3CA inhibitors (3)	**2017** PIK3CA-mutant cell-line panel (CCLE) and PIK3CA-mutant breast cancer xenografts	**2015** ER+ PIK3CA-mutant breast lines, xenografts and PDX under… No checkable prediction	**2020** Index patient's PTEN-null, BYL719-refractory metastasis PDX; PTEN-knockdown PIK3CA mutants	**2015** ER+ PIK3CA-mutant breast lines, xenografts and PDX under… No checkable prediction
	breast cancer pan-HER inhibitors (3)	**2014** Tamoxifen-resistant MCF7/T47D xenografts Not measured in patients	**2017** HER2+ breast lines and mice	**2017** HER2+ breast lines and mice	**2017** HER2+ breast lines and mice
#10	**osteoarthritis** MAPK14 inhibitors (3)	**2012** Rat iodoacetate joint degeneration + carrageenan thermal hyperalgesia (SB-203580, VX-745)	**2009** Rat and cyno endotoxin TNF-alpha PD Patient result unclear	**2012** Rodent rheumatoid-arthritis models, ex-vivo LPS whole-blood cytokine PD (ARRY-797)	**2014** Rat repeat-dose intra-articular toxicology (FX005) Not measured in patients
	osteoarthritis NTRK1 inhibitors (3)	**2019** Rat MIA and MNX knee OA pain model (single-drug… Not measured in patients	**2019** Rat MIA and MNX knee OA pain model (single-drug… No checkable prediction	**2019** Human NTRK1 loss-of-function (congenital… △ ASP7962 failed WOMAC pain…	**2019** Rat naive-knee intra-articula… Rat silent; two-thirds got joint inflammation
#11	**colorectal cancer** KDR (VEGFR) inhibitors (3)	**2014** Human tumour cell-line xenografts in nude rodents (incl. VEGF-challenge pVEGFR2 readout)	**2014** Human tumour cell-line xenografts in nude rodents (incl.… △ Start dose toxic; animal margin…	**2017** Human colorectal patient-derived xenografts (PDX) i… △ BATON-CRC PFS futility vs…	**2023** Rat, dog and monkey… Dogs silent; 14% grade 3-4 hypertension
	colorectal cancer VEGFR multikinase inhibitors (3)	**2007** VEGF-induced uterine edema in mice (vascular-permeability PD model, ABT-869/linifanib)	**2007** Human tumour cell-line xenografts in mice incl. colon carcinoma (ABT-869/linifanib)	**2015** Human tumour cell-line xenografts in mice incl. colon… △ Single-arm: 0/30 responses,…	**2014** Rodent linifanib hypertension Hypertension in patients as rats predicted
	colorectal cancer anti-EGFR antibodies (3)	**2018** EGFR-expressing cell lines and xenografts incl. cetuximab-resistant and…	**2017** Resistant CRC lines No checkable prediction	**2017** Resistant CRC lines Phase 2 OS failed in acquired resistance	**2013** Cynomolgus repeat-dose toxicology with cetuximab… △ Well tolerated in mice; not in…
#12	**chronic hepatitis B virus infection** HBV RNA antisense / siRNA (3)	**2020** HBV-genome mice (pHBV transient/transgenic, later AAV-HBV)	**2018** HBV-infected human liver-chimeric mice No checkable prediction	**2024** HBV-genome mice (pHBV transient/transgenic, later AAV-HBV)	**2026** HBV-genome mice (pHBV transient/transgenic, later AAV-HBV) No ALT rise in mice, as in volunteers
	chronic hepatitis B virus infection HBsAg release (NAP) inhibitors (3)	**2017** DHBV-infected Pekin ducks	**2017** CD-1 mouse and cynomolgus PK/toxicology (uninfected)	**2017** DHBV-infected Pekin ducks REP 2165 arm matched, after the fact	**2017** Primary human hepatocytes… Human cells reproduce weak immunostimulation, partly
	chronic hepatitis B virus infection TLR7 agonists (3)	**2013** Chronically HBV-infected chimpanzees	**2015** WHV-infected woodchucks No checkable prediction	**2016** WHV-infected woodchucks ISG-matched dosing, no HBsAg effect	**2015** Chronically HBV-infected chimpanzees △ Patients: no thrombocytopenia a…
#13	**hepatocellular carcinoma** VEGFR multikinase inhibitors (4)	**2015** Human HCC cell-line and PDX xenografts Not measured in patients	**2015** Human HCC cell-line and PDX xenografts	**2020** Mouse HCC with diverse transposon-delivered genetic drivers, lenvatinib vs sorafenib susceptibility	**2018** GLP repeat-dose toxicology in rat plus dog or monkey
	hepatocellular carcinoma anti-PDCD1 antibodies (4)	**2015** Syngeneic orthotopic mouse HCC under a rat anti-mouse PD-1… Not measured in patients	**2025** Population-PK simulation of… Simulated trough 20x occupancy target; APOLLO met	**2022** Syngeneic orthotopic mouse HCC under a rat anti-mouse PD-1 surrogate, with or without apatinib	**2022** PD-1 knockout mice △ Monkeys silent; patients had irAEs
	hepatocellular carcinoma KDR (VEGFR) inhibitors (3)	**2020** Human liver-cancer xenografts, rats and mice	**2021** Human liver-cancer xenografts, rats and mice	**2020** Human liver-cancer xenografts, rats and mice Tivozanib HCC trial stopped early	**2020** GLP repeat-dose tox and cardiovascular safety pharmacology in rodent plus dog or monkey
	hepatocellular carcinoma anti-CD274 antibodies (3)	**2020** Primary human T-cell activation assay (proliferation, IFN-gamma) with durvalumab	**2021** Cynomolgus monkey 4-week and 13-week GLP repeat-dose toxicology of durvalumab	**2021** Syngeneic CT26 mouse colorectal carcinoma dosed with the rat anti-mouse PD-L1 surrogate…	**2021** Cynomolgus monkey 4-week and 13-week GLP repeat-dose toxicology of durvalumab
#14	**schizophrenia** PDE10A inhibitors (5)	**2019** Rat and monkey ketamine-challenge phMRI/EEG reversal	**2013** Rodent occupancy dosing	**2013** Rodent CAR / psychostimulant hyperlocomotion Monotherapy PANSS not met	**2017** Rat catalepsy, prolactin and glucose vs D2 antagonists EPS in patients at clinical dose
	schizophrenia SLC6A9 inhibitors (4)	**2016** Rat CSF/striatal glycine by microdialysis △ Ketamine WM readout unmoved	**2023** Human visual-evoked LTP in schizophrenia	**2023** Human ketamine-challenge working-memory fMRI Assay said inactive; trial flat	**2026** Rodent haematology (haemoglobin) in pharmacology and toxicology
	schizophrenia CHRNA7 agonists (3)	**2024** Pig [11C]NS14492 PET alpha7 occupancy Not measured in patients	**2011** Rodent screens: PPI, social behaviour, object recognition No checkable prediction	**2015** Rodent screens: PPI, social behaviour, object recognition ABT-126 phase 2b missed MCCB	**2011** Rodent screens: PPI, social behaviour, object recognition
#15	**gastric cancer** PARP1/2 inhibitors (3)	**2023** Gastroesophageal PDX, BRCA2-mutant, PARP inhibitors Not measured in patients	**2023** Gastroesophageal PDX, BRCA2-mutant, PARP inhibitors No checkable prediction	**2026** Gastroesophageal PDX, BRCA2-mutant, PARP inhibitors △ Pamiparib maintenance missed…	**2017** Rat and dog general toxicology, up to 26 weeks (olaparib)